

Extending the MultiLexNorm++ Pipeline with Length-Aware Detection and Language-Specific Threshold Optimization

Jiayang Zhang, Yujiaxuan Wang
Advisor: Kathy Hämmerl

NLP & LLMs Practical Course, Technical University of Munich

Introduction

Lexical Normalization

Raw: *Why do dese guys think they doin' summn?*
Normalized: *Why do these guys think they are doing something?*

MultiLexNorm++

- A benchmark with origin Indo-European and new Asian languages
- A detector-dictionary-LLM normalization pipeline

Challenge

- Binary detection ignores output length
- One fixed thresholds setting may not be optimal for all languages

Our work

Investigate length-aware detection and language-specific threshold optimization

Method

Reproduced Baseline

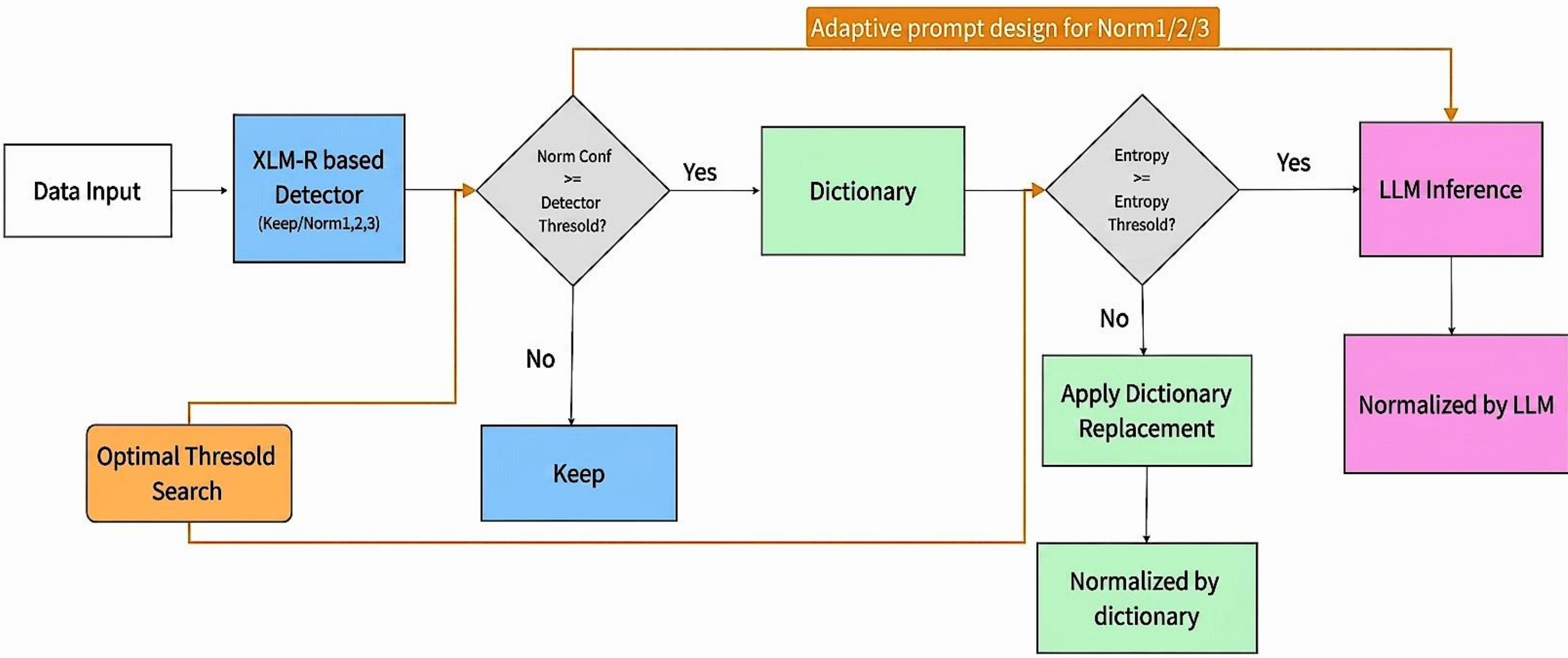
- Reproduced the original pipeline (blue – green - pink)

Our Pipeline – Fine-grained Detector

- Train length-aware detector to replace original binary classifier
- Original Label: KEEP/NORM New Label: KEEP/ NORM_1/2/3+
- Normalization: SUM(NORM_1/2/3+) >= detector threshold
- Build prompts with length-constraint for LLM inference

Our Pipeline – Search Optimal Thresholds as Hyperparameters

- Grid-searched detector and entropy thresholds per language
- First run with lowest thresholds to maximize LLM candidates
- Cache first run outputs and reuse them for later combinations
- Select the combination with best ERR on the DEV



Experimental Setup

Dataset

- MultiLexNorm++: 7 Indo-European languages + 5 Asian languages
- Training data: 35.9k rows, split into 90% TRAIN and 10% DEV
- Validation data: 8.4k rows, used for final evaluation

Models

- Detectors: XLM-R trained with MaChAmp setup
- LLMs: Qwen3.5:9b and Deepseek-V4-Pro

Evaluation Metrics

- ERR: how much pipeline improves over "Do Nothing"?
- F1: does pipeline keep balance between precision and recall?

Thresold Grid Search

- Detector Thresold Pool: {0.1, 0.3, 0.5, 0.7, 0.9}
- Entropy Thresold Pool: {0.2, 0.5, 0.8, 1.1, 1.4, 1.7}

Results

Overall Performance

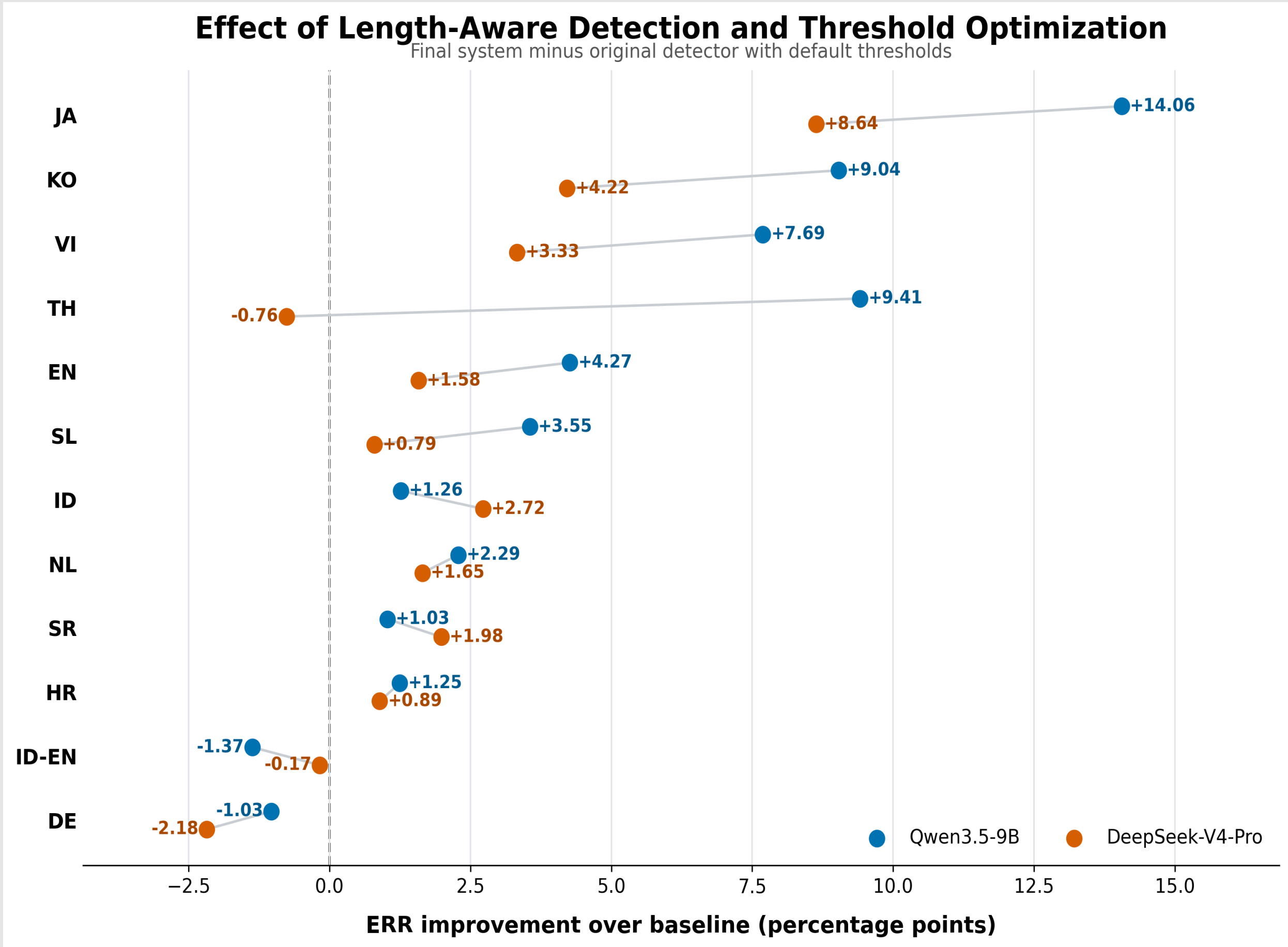
- The full optimized pipeline improves slightly
- New pipeline achieves positive overall gains in 9/12 languages
- Optimized threshold search consistently improves overall performance

Macro-average Validation Results across 12 Languages (%)

Configuration	Qwen ERR ↑	Qwen F1 ↑	DeepSeek ERR ↑	DeepSeek F1 ↑
Original detector + default thresholds	43.61	64.26	54.52	72.48
Length-aware detector + default thresholds	43.14	63.35	55.09	72.50
Length-aware detector + optimized thresholds	47.90	67.01	56.41	73.35
Final improvement over baseline	+4.29	+2.75	+1.89	+0.88

Improvement on Asian vs. Indo-European Languages

- New pipeline yields larger gain for Asian languages than non-Asian
- Benefits are not uniform across all Asian languages



Ablation: Length-aware Detector under Default Thresholds

- With new detector alone, 4 out of 5 Asian languages improves in ERR

lang. / group	Qwen ΔERR	Qwen Δf1	DeepSeek ΔERR	DeepSeek Δf1
ja	+2.64	+0.19	+1.76	+1.09
ko	+3.01	-6.40	+2.41	-5.58
th	-7.12	-0.82	-2.54	+1.41
vi	+0.89	+0.60	+0.61	+0.42
id	+1.02	+0.63	+2.24	+1.50
non-Asian	-0.88	-0.73	+0.34	+0.20

Conclusion & Limitation

Conclusion

- Length-aware detection alone is mixed overall, but more often helps Asian languages. Optimal results are achieved after threshold search
- Final pipeline improves most languages and benefits Asian more

Limitations

- Test labels are not publicly available, so pipeline relies on DEV set
- Some languages still suffer from false positive predictions, likely due to limited train data and ambiguous normalization boundaries
- ERR of Asian languages are still lower than that of non-Asian

References

[1] ÚFAL at MultiLexNorm 2021: Improving Multilingual Lexical Normalization by Fine-tuning ByT5[https://aclanthology.org/2021.wnut-1.54/] (Samuel & Straka, WNUT 2021)
[2] Buaphet, W., Nguyen, T. N., Kondo, R., Kajiwar, T., Kim, Y., Lee, J., ... & Van der Goot, R. (2026). MultiLexNorm++: A Unified Benchmark and a Generative Model for Lexical Normalization for Asian Languages.